TECHNICAL ARCHITECTURE & BUSINESS PROPOSAL

Call Recording Implementation Strategy

Feasibility Study, Architecture Comparison & Implementation Proposal for Hi Talk CRM

APPLICATION	PREPARED FOR	DATE & VERSION	STATUS
Hi Talk Telecalling CRM	Management & Approvals	October 14, 2026 | v1.0	Pending Management Review

1 Executive Summary

This report provides a comprehensive technical evaluation and financial feasibility study for enabling **Call Audio Recording** across the **Hi Talk CRM Mobile Application**. Call recording is essential for our telecalling operations to ensure call quality monitoring, agent sales pitch training, customer dispute resolution, and regulatory compliance.

> **Key Finding:** Modern mobile operating systems (Android 10 through Android 15) strictly disallow third-party applications from directly tapping into cellular voice streams via internal audio APIs. In response, enterprise CRM systems utilize one of two industry-proven models: **Option A (Local Auto-Sync via Existing Infrastructure at ₹0 extra cost)** or **Option B (Cloud Telephony Integration via Exotel with per-minute billing)**.

2 Technical Landscape & Operating System Constraints

To make an informed decision, it is critical to understand why standard mobile apps cannot simply record phone calls out-of-the-box:

- **Google Android Policy & Privacy Restrictions (Android 10 - 15):** Starting in Android 9 and fully enforced in Android 10+, Google deprecated and blocked the `VOICE_CALL` audio stream for third-party apps. Regular apps cannot access incoming caller audio during cellular calls.
- **Microphone Exclusivity:** When a phone call is connected, the Android Telecom Subsystem claims exclusive priority over the microphone hardware. Conventional audio recorder packages (like `react-native-audio-recorder-player`) will fail, crash, or capture only silence.
- **Play Store Compliance:** In May 2022, Google barred the use of the Accessibility API for call recording purposes.

To achieve reliable audio capture, industry leaders such as *Runo, LeadSquared, and TeleCRM* rely on the two viable strategies detailed below.

3 Architectural Options for Hi Talk

RECOMMENDED FOR PHASE 1

Option A: Local Call Recording Sync

Option B: Cloud Telephony (Exotel)

Native Phone Recording + Automatic Server Upload

₹0 Recurring Telecom Cost (100% Free)

Agents utilize the built-in "Auto Call Recording" feature available in Android handsets. Upon call completion, the Hi Talk app automatically identifies the audio file and uploads it to our Hostinger Server.

Key Advantages:

- Zero monthly subscription or per-minute call charges.
- 100% crystal-clear two-sided audio (Caller + Agent).
- Utilizes existing Hostinger VPS/SSD storage without extra servers.
- Rapid 2-day implementation lifecycle.

Limitations:

- Requires telecallers to enable "Auto Call Record" once in device settings.
- Telecallers place calls from their physical SIM cards (no number masking).

Virtual Number Bridge & Centralized Cloud Recording

Paid (₹1,500/mo + ₹0.75/min)

When the agent taps "Call", Exotel's cloud telephony servers bridge the call between the agent and customer. The call is recorded entirely on Exotel's servers and streamed to our CRM via webhooks.

Key Advantages:

- **Number Masking:** Protects lead numbers; agents cannot harvest customer contacts.
- Single corporate Caller ID builds brand trust and minimizes spam flags.
- 100% device independent (works on Android, iOS, and landlines).
- Live call listening and manager whispering capabilities.

Limitations:

- Recurring monthly bill + per-minute call pulse costs.
- Mandatory telecom paperwork (GST, company KYC, DLT registration).

4 Deep Dive: Option A Workflow & Storage Feasibility

Option A transforms the existing Android architecture into an automated recording sync pipeline with zero third-party software cost:

1. **Native Call Recording:** Agent initiates a call via Hi Talk. The device's native dialer (Samsung, Xiaomi, Realme, Vivo, Oppo) automatically records both voices to local internal storage (`/Recordings/Call`).
2. **Call Termination Detection:** The existing `android-call.service.ts` in our mobile app captures call end events and verifies the call duration.
3. **Background File Identification & Upload:** The app immediately queries the recording folder for the newest audio file matching the lead's phone number and initiates a background multipart upload to `POST /api/leads/:id/recordings`.
4. **Admin CRM Playback:** The Hostinger server links the audio file to the lead history. Administrators and managers can stream or download the call recording directly from the Admin Portal.

Storage Impact Analysis on Current Hostinger Server

Standard mobile call recordings are encoded in compressed `.m4a` or `.aac` format at 32 kbps, requiring negligible disk space:

Metric	Single Agent	5 Agents Team	10 Agents Team
Daily Call Duration	100 minutes / day	500 minutes / day	1,000 minutes / day
Daily Storage Consumption	~24 MB	~120 MB	~240 MB
Monthly Storage (25 Days)	~600 MB	**~3.0 GB**	**~6.0 GB**
Annual Storage Impact	~7.2 GB	~36 GB	~72 GB

Infrastructure Feasibility: Our Hostinger Server plan includes substantial SSD storage capacity. With a 5-agent team, an entire year of continuous call recordings consumes approximately **36 GB**. This fits comfortably within our current hosting quota without purchasing additional storage.

5 Comprehensive Feature & Operational Comparison

Evaluation Parameter	Option A: Local Sync (In-House)	Option B: Exotel Cloud Telephony
Telephony Setup Cost	**₹0** (Zero)	₹1,500 – ₹3,000 (One-time + Virtual DID)
Recurring Monthly Cost	**₹0 / month**	₹1,500/mo DID rent + call usage
Outgoing Call Rate	Free (Existing Unlimited SIM Plans)	₹0.60 – ₹0.85 per minute

Evaluation Parameter	Option A: Local Sync (In-House)	Option B: Exotel Cloud Telephony
Audio Storage Location	Hostinger Server (Existing SSD)	Exotel Cloud & Hostinger Webhooks
Admin Remote Playback	Yes (Instant stream via Admin Portal)	Yes (Instant stream via Admin Portal)
Number Masking (Lead Privacy)	No (Lead number visible on dialer)	**Yes (Complete Masking / Protection)**
Agent Device Setup	Enable Auto-Record in Phone Settings	Zero setup (Works on any handset)
Handset Compatibility	Samsung, Xiaomi, Realme, Vivo, Oppo	100% all devices (Android, iOS)
Inbound IVR & Call Routing	Not available	Yes (Multi-level IVR greeting)
Time to Production	**2 to 3 Business Days**	10 to 14 Days (due to KYC & DLT)

6 12-Month Financial Cost Forecast (Team of 5 Telecallers)

The cost projection assumes 5 telecallers, each handling 40 connected calls/day with an average duration of 2 minutes (400 minutes/day $\times$ 25 days = **10,000 minutes/month**):

Expense Component	Option A (Local Sync on Hostinger)	Option B (Exotel Cloud Telephony)
Virtual DID Numbers (1 Line)	₹0	₹24,000 / year (₹2,000/mo)
Call Minute Charges (120,000 mins/yr @ ₹0.75)	₹0 (Covered by agent SIM unlimited plans)	₹90,000 / year
Server Storage & Maintenance	₹0 (Covered under current Hostinger plan)	₹0
Implementation / Setup Fee	In-house development	₹3,000 (Account onboarding)
Total Estimated Annual Expenditure	**₹0 / Year**	**₹1,17,000 / Year**

7 Strategic Recommendation & Roadmap

> **Strategic Consensus: A Two-Phase Rollout Plan**
>
> Given the budget parameters and operational readiness of our team, we recommend a **phased approach** that delivers immediate value while providing a future upgrade path.

Phase 1: Immediate Implementation (Option A)

Implement **Local Call Recording Sync** immediately on the current Hostinger infrastructure.

- Immediate cost savings: saves **₹1.17 Lakhs** in year one.
- Provides instant call monitoring and QA playback for managers.
- Go-live achievable within **2 to 3 days**.

Phase 2: Enterprise Transition (Option B)

Upgrade to **Exotel Cloud Telephony** when the business scales to the following trigger points:

- Telecaller team grows beyond 15–20 active agents.
- Lead data poaching / customer number masking becomes legally mandatory.
- Requirement for corporate toll-free numbers and IVR customer routing.

8 Action Requested & Management Approval

Kindly review the proposed options and signify approval to initiate development:

Management Decision:

- ☐ **[APPROVED] Option A:** Implement In-House Local Recording Sync (Hostinger Server - ₹0 cost).

☐ **[APPROVED] Option B:** Proceed with Exotel Cloud Telephony Integration (Procure DID & API credentials).

☐ **[REVISE]:** Request additional details / schedule review meeting.

Prepared By: Engineering Team (Hi Talk CRM)
Date: October 14, 2026

Approving Manager:

Signature & Date: _____